

EMPLOYEE ENGAGEMENT, SATISFACTION, AND BURNOUT DIAGNOSTIC ANALYSIS

@ PALO ALTO NETWORKS

Data Sources :

Project Field : Data Analytics

Project Name : Palo Alto Networks

Tools Used : Python, Pandas, Numpy, Matplotlib, Seaborn, Plotly, Streamlit

Data Source : Palo Alto Networks.csv

Abstract :

In modern cybersecurity organizations, employee engagement, satisfaction, and burnout directly influence productivity, innovation, retention, and organizational performance. Sometimes high-pressure environments, continuous operational demands and rapid technological changes increase the chances of employee stress and emotional exhaustion. Employee burnout and disengagement are key factors that lead to attrition. Factors such as workload, work-life balance, job satisfaction, overtime frequency, performance ratings and career growth opportunities are analyzed to identify burnout-prone employees. The project focuses on supporting strategic HR decision-making by enabling proactive interventions, improving employee well-being, and reducing organizational turnover costs. These valuable findings show how data-driven HR analytics can enhance employee experience and organizational sustainability in high-performance technology companies.

Introduction :

This is a dataset of 1470 employees of Palo Alto Networks which operates in a high performance environment where employee efficiency and innovation are critical for maintaining leadership. This project develops an employee diagnostic system for Palo Alto Networks that helps HR teams to identify key factors like employee engagement, satisfaction, work-life balance, burnout risk. Employee engagement and workplace well being have become major priorities for organizations. Organizations therefore require diagnostic system capable of identifying risk factors before employees decide to leave(attrition).

Problem Statement :

Palo Alto Networks currently lacks:

- A unified engagement health index
- Clear visibility into burnout-prone employee groups
- Understanding of how overtime, work-life balance, and satisfaction interact
- Preventive insights into employee experience deterioration

Objectives :

The main objectives of this project research are :

- To analyze employee engagement and satisfaction patterns.
- To identify factors contributing to employee burnout.
- To develop a predictive model for burnout risk classification.
- To evaluate the relationship between workload and employee satisfaction.
- To provide HR-driven recommendations for improving workforce well-being.
- To support proactive employee retention strategies.

Dataset Overview :

1470 Rows * 31 Columns The dataset contains employee-related attributes such as:

Age, Department, Job role Monthly income, Work-life balance, Overtime frequency, Job satisfaction, Performance rating, Years at company, Training frequency, Engagement score, Burnout level

Methodology :

1. Data Cleaning & Processing :

- Removal of duplicates
- Removal of Invalid records
- Removal of null values
- Handle missing values
- Standardize label names

2. Creating KPI (Key Performance Indicator) :

We have to create some important KPIs :

- Engagement Index : Composite score of involvement & satisfaction
- Burnout Risk : Overtime + low work-life balance indicator
- Work-life Balance : Average balance rating

- Satisfaction stability score : Consistency of satisfaction across dimensions
- Workload stress indicator : Travel + overtime intensity

3.Exploratory Data Analysis :

Data is grouped by Burnout Risk, Attrition, JobRole, Engagement Index, and YearsSinceLastPromotion .

Techniques used:

- Correlation heatmaps
- Histograms
- Box plots
- Pie charts
- Burnout distribution analysis
- Department-wise satisfaction analysis

4. Data Visualization :

Data is visualized by different categories like impact of Business Travel, department, age, education field, gender and overtime on Attrition. Engagement index by overtime Vs non-overtime , Attrition rate % by Burnout Risk Level, Engagement index distribution.

5. Creating Dashboard :

Create an attractive, interactive dashboard which helps decision-making easier.

Streamlit App Dashboard :

Dashboard  :

<https://ckfudpmga2hqn6hzigpzyp.streamlit.app/>

Findings / Insights :

1. Organization-Wide Baseline: Out of 1,470 analyzed employees:

- **Baseline Attrition Rate:** 16.12%
- **Average Engagement Index:** 57.44%
- **Average Satisfaction Stability:** 45.25% (indicates substantial imbalance, with many employees having at least one critical dissatisfaction dimension).
- **Burnout Risk Distribution:**
 - **Low Risk:** 51.43% (756 employees)
 - **Medium Risk:** 40.00% (588 employees)
 - **High Risk:** 8.57% (126 employees)

2. Burnout as a Primary Driver of Attrition

The data reveals a stark, linear correlation between our computed Burnout Risk and actual historical attrition:

Burnout Risk Category	Active Employees	Leaver Employees	Attrition Rate (%)
Low	687	69	9.13%
Medium	462	126	21.42%
High	84	42	33.33%

Insight: Employees in the High Burnout Risk category leave the organization at **more than three times** the rate of Low Risk employees. Over 33% of high-risk employees exit, demonstrating the high accuracy of the Burnout Risk Classifier as an early warning metric.

3. Risk Concentration by Department and Job Role

Burnout and disengagement are not distributed evenly. R&D and Sales bear the brunt of work overload.

- **Departmental Burnout Rates:**
 - **Research & Development:** 8.84% High Burnout Risk
 - **Sales:** 8.74% High Burnout Risk
 - **Human Resources:** 3.17% High Burnout Risk
- **Top Roles by Burnout Concentration:**
 1. **Research Scientists:** 10.96% High Burnout Risk
 2. **Laboratory Technicians:** 9.27% High Burnout Risk
 3. **Sales Executives:** 8.90% High Burnout Risk
 4. **Research Directors:** 8.75% High Burnout Risk

4. Workload and Stress Analysis

We evaluated how external variables—namely business travel and commute distances—interact with engagement.

- **Commute and Attrition:**
 - Short-commute employees (< 10 miles): **14.15% attrition rate**
 - Long-commute employees (>= 10 miles): **19.62% attrition rate**
- **Travel Intensity:** Employees who travel frequently maintain a higher Workload Stress average (**75% to 100%** on our scale when paired with overtime), significantly increasing their risk profile.

5. Career-Stage & Stagnation Analysis

We analyzed engagement across job levels and tenures to identify disengagement trigger points.

- **The Mid-Level Dip:** Engagement drops at **Job Level 3 (55.81%)** before rebounding at **Job Level 4 (59.43%)**. This suggests that mid-level individual contributors or

managers feel squeezed between senior leadership demands and junior execution support.

- **Tenure Milestones:**
 - At **Years Since Last Promotion** = 6, average engagement drops to **53.91%**.
 - At **Years In Current Role** = 3 to 4, engagement dips to **~55.7%**.
 - At **Years With Current Manager** = 3, engagement drops to **55.58%**, indicating that relationships can stagnate or sour over time if not refreshed.
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Recommendations :

- Maintain healthy work environment
- Prioritize employee emotional and mental well-being
- Prioritize those employees who are at Burnout Risk and ready for attrition
- Reduce overtime work
- Engage and interact with every employee

Conclusion :

Employee engagement and burnout management are critical challenges in high-performance organizations. This research proposes a data-driven diagnostic framework capable of identifying early burnout risks and satisfaction patterns among employees. By applying HR analytics and machine learning techniques, the project supports proactive organizational decision-making and employee well-being initiatives.

The study demonstrates that predictive analytics can significantly improve workforce management by helping organizations identify hidden engagement issues before they result in attrition. The proposed framework can help organizations like Palo Alto Networks build healthier, more productive, and sustainable work environments.

Project Links :

Github Repository : <https://github.com/Nivedita-1998/Employee-Engagement-Satisfaction-and-Burnout-Diagnostic-Analysis-Dashboard--Palo-Alto-Networks/blob/main/requirements.txt>

Streamlit App Dashboard :
<https://ckfudpmga2hqn6hzigpzyp.streamlit.app/>